

$$\mathcal{U}_1 = \begin{array}{c}
 |p_n\rangle \text{ --- } \boxed{R_Z(m\varepsilon)} \text{ ---} \\
 |\sigma_{n,n+1}\rangle \text{ --- } \boxed{R_X(\varepsilon)} \text{ ---} \\
 |e_{n+1}\rangle \text{ --- } \boxed{R_Z(-m\varepsilon)} \text{ ---}
 \end{array}$$